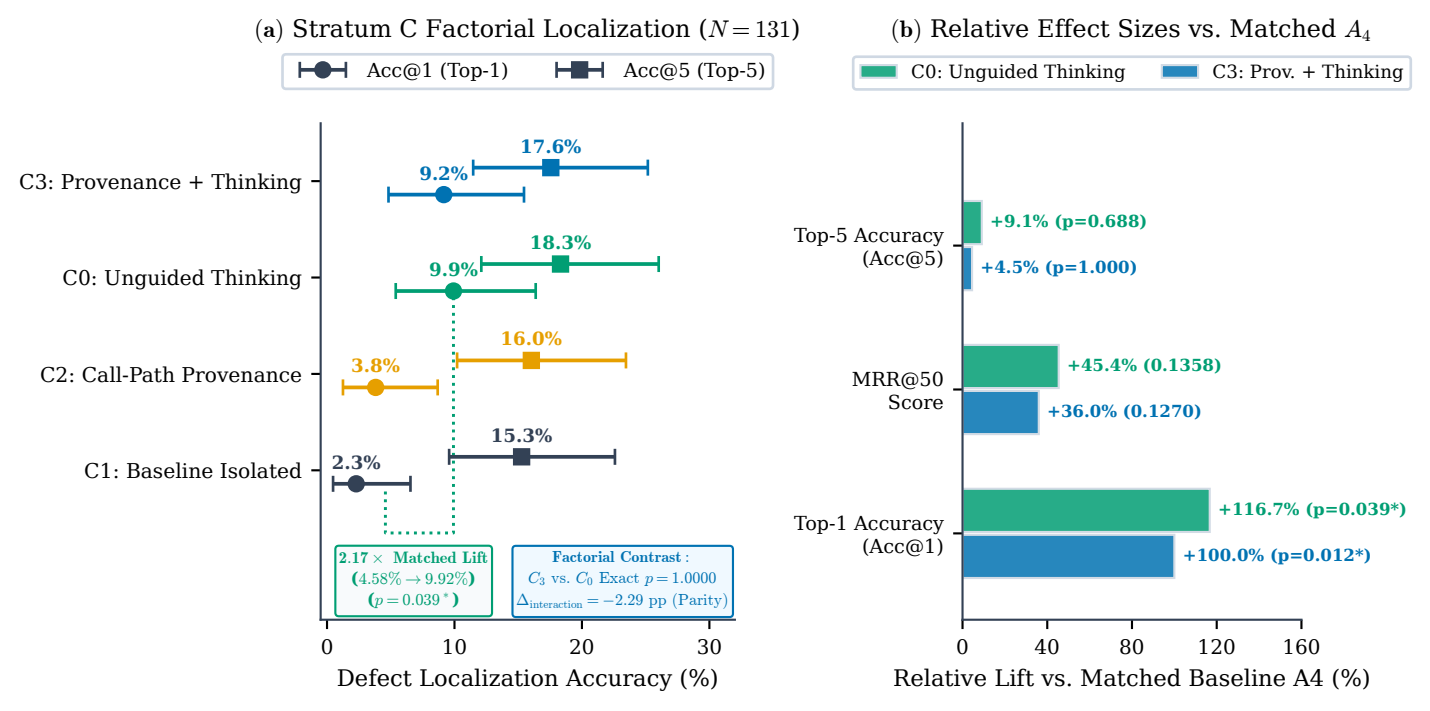
(a) Stratum C Factorial Localization ($N = 131$)
Acc@1 (Top-1)
Acc@5 (Top-5)
C3: Provenance + Thinking
C0: Unguided Thinking
C2: Call-Path Provenance
C1: Baseline Isolated
17.6%
9.2%
18.3%
9.9%
16.0%
3.8%
15.3%
2.3%
2.17× Matched Lift
(4.58% → 9.92%)
($p = 0.039$ *)
Factorial Contrast :
C_3 vs. C_0 Exact $p = 1.0000$
$\Delta_{\text{interaction}} = -2.29$ pp (Parity)
0
10
20
30
Defect Localization Accuracy (%)
(b) Relative Effect Sizes vs. Matched A_4
C0: Unguided Thinking
C3: Prov. + Thinking
Top-5 Accuracy
(Acc@5)
MRR@50
Score
Top-1 Accuracy
(Acc@1)
+9.1% (p=0.688)
+4.5% (p=1.000)
+45.4% (0.1358)
+36.0% (0.1270)
+116.7% (p=0.039*)
+100.0% (p=0.012*)
0
40
80
120
160
Relative Lift vs. Matched Baseline A4 (%)